

T 37/82 (OJ 1984, 71) The board held that in assessing the inventive step of a combination of features, consideration had to be given to a feature only if the applicant had provided evidence that it contributed, either independently or in conjunction with one or more of the other features, to the solution of the problem set in the description (see also T 65/87, T 144/90, T 206/91, T 226/94, T 912/94, T 15/97, T 471/98, T 442/02). Therefore, only those claimed features are to be considered which contribute causally to the solution of the problem (T 285/91). In T 294/89 the board stated that the additional feature provided no surprising advantage and did not make any contribution to solving the problem indicated. Hence, the said additional feature was not relevant for assessing the inventive step of the combination of features claimed.

In T 589/95 the terms of the solution of the technical problem extended into an area of use where it had been admitted that the relevant problem was known not to arise in practice. The board stated that, for such an area, the features of the solution did not contribute to the solution of the technical problem and could not be taken into account in the assessment of inventive step.

With reference to T 119/82, Board 3.3.05 held in T 72/95, T 157/97, T 176/97 and T 158/97 that similar considerations applied to technically non-functional modifications. An inventive step could not be claimed on the basis of a non-functional modification of a known device. If a known device was modified by adding a feature which had no technical function, this modification could not be inventive (see also in this chapter I.D.9.1. "Assessment of inventive step in the case of mixed-type inventions").

In T 1009/12 the board took the view referring to T 206/91 that an ineffective concentration of a compound was considered as an arbitrary feature not contributing to the solution of the underlying problem and therefore not further considered. Since there were no further distinguishing features it was not feasible to identify the technical problem to be solved and the claim lacked inventive step.

In T 2044/09 the board held that even if there was no pointer or suggestion in the prior art towards the addition of a distinguishing feature, if said modification was not linked to a particular functionality, then it could not per se constitute the basis for acknowledging an inventive step.

In T 2287/16 the board had no objection to the examining division ignoring, in an inventive-step assessment, a feature that it found to contravene Art. 123(2) EPC. The board considered that the examining division was not obliged to anticipate the replacement of such a feature with a similar one that might have been originally disclosed and to provide a speculative inventive step assessment of a so-amended claim.

## 9.7. Substitution of materials – analogous use

In T 21/81 (OJ 1983, 15) the board considered it as forming part of his normal activities for a skilled person to select, from the materials known to him as suitable for a certain purpose, the one which was the most appropriate. In T 324/94 the board held that the skilled person should therefore be at liberty, within the constraints of standard technical progress, to use

alternative means known by him to have the same effect. In T.410/92 the board concluded that the skilled person who encountered known problems when using newly developed materials would not be deterred from using them in order to achieve specific, desired improvements, particularly since the means of overcoming such problems could be derived from the prior art.

The Headnote in T.192/82 (OJ 1984, 415) reads as follows: If an article is known as a combination or mixture of components fulfilling known functions, the generation and application of an improved novel component for the same purpose may be patentable as such and also as an improved article incorporating the same. If the component in question forms, on the other hand, part of the state of the art together with its relevant properties, the incorporation thereof in the same article will be obvious in view of its predictable beneficial effect ("analogous substitution").

In this regard the board in T.130/89 (OJ 1991, 514) held that that the use of a known material in a combination on the basis of its known properties and in a known manner to obtain a known effect was not normally inventive ("similar use"). Exceptions to this principle might be allowed in special cases, e.g. where a selection brought unexpected advantages, a known prejudice was overcome or unforeseen difficulties were encountered, such as the need to alter another component (see also T.1216/05, T.330/07, T.422/09).

Following these decisions, the board summarised as follows in T.213/87; in the absence of any unexpected effect, the mere substitution of an element by another known for its relevant properties to provide that known effect could not be regarded as patentable.

## 9.8. Combination of documents

It would not be obvious to a skilled person to combine an isolated, **very old document**, which had not given rise to a trend in the art and whose teaching ran counter to the present trend, with the document reflecting the closest state of the art. See for example T.261/87 (literature from 1907), T.366/89 (a document filed in 1927), T.404/90 (a "relatively old" document published in 1949).

In T.278/14 the board rejected the argument that the need to combine three pieces of prior art in order to arrive at the claimed process confirmed the presence of an inventive step. The board held that there was no general principle which determines that in order for a lack of inventive step to be established only two documents may be combined.

In T.745/92 the board pointed out that the disclosure of two prior documents – even if they were classified under the same IPC classification – could only be combined so as to result in a finding of lack of inventive step if such combination would have been obvious to a skilled person seeking to solve the problem underlying the claimed invention (see also T.104/95, T.395/00).

In T.715/09 the board stated that IPC classification alone was no reason for determining whether or not two pieces of prior art could be combined. The mere fact that two

documents had the same classification was no reason for saying the combination of the teachings was obvious (T. 745/92). Likewise the mere fact that the technologies had been given different IPC classes did not necessarily mean that they could not be combined.

In T. 454/13 the invention concerned a sample block apparatus and a method for maintaining a microcard containing a plurality of samples of biological material on a sample block. The problem to be solved consisted in improving temperature uniformity between the plurality of samples on a single microcard. According to appellant 2 (opponent), the solution was obvious in view of a combination of document D1 with any of documents D6 to D8. The board found that as the issue of biological samples was not normally encountered in the field of semiconductor manufacturing, the skilled person would have no incentive to look for a solution in that field. The board concluded that the skilled person would therefore have been dissuaded from consulting documents D6 to D8.

In T. 552/89 the board confirmed that, when assessing inventive step, it was not permissible to combine the teachings of different documents within the state of the art in order to establish the obviousness of a claimed invention, unless it would have been obvious for the skilled person to do so at the time of filing. When a problem defined by reference to the closest prior art as disclosed in a primary document consisted of individual problems, the case law of the boards of appeal stated that the skilled person could be expected to take account of solutions to the individual problems proposed in different secondary documents in the same or neighbouring technical fields. Thus, the teachings of secondary documents might be combined with the disclosure of the closest prior art if such secondary documents provided solutions to specific individual problems forming part of the objective problem in progressing from the closest prior art, in particular when such individual solutions were merely aggregated together in the claimed invention (see also T. 302/02, T. 881/09).

In T. 302/02 the board added that, if an invention consisted of a new combination of features taken from different technical areas, a discussion whether or not it was obvious would normally involve at least as many documents as technical areas combined in it. The board adopted the approach taken in T. 552/89, according to which a technical problem might be formed of "individual problems". The number of individual problems obviously depended on the degree of detail of the claim under consideration and the cited decision did not suggest that beyond a certain number the presence of an inventive step might be taken for granted. On the contrary, it was said to be obvious to try to solve the individual problems as long as the corresponding solutions were "merely aggregated together" in the claim.

In T. 881/09 the board stated that since the teachings of both documents stood in heavy discrepancy to each other, in view of their incompatibilities a combination of these documents was not obvious (T. 552/89) but rather artificial and could only be the result of an ex post facto analysis.

## 9.9. Chemical inventions

### 9.9.1 Problem and solution approach in chemical inventions

In a number of chemistry decisions the problem and solution approach – based on landmark decisions T.1/80 (OJ 1981, 206), T.24/81 (OJ 1983, 133) and T.248/85 (OJ 1986, 261) – involves the following steps:

- a) establishing the closest prior art;
- b) defining the problem in the light of that prior art;
- c) identifying the solution;
- d) demonstrating the success of the solution;
- e) optionally reformulating the problem;
- f) examining the obviousness of the solution in view of the state of the art.

Demonstrating the success of the solution and reformulating the problem are particularly important steps (see T.231/97, T.355/97).

### 9.9.2 Structural similarity

In T.852/91 the board held that to deny inventive step for novel chemical compounds because of their structural similarity to known chemical compounds amounted to an allegation that a skilled person would have reasonably expected the same or similar usefulness of both the known and the novel compounds as the means for solving the technical problem underlying the application in question. Such an expectation would be justified, if the skilled person knew, be it from common general knowledge or from some specific disclosure, that the existing structural differences of the chemical compounds concerned were so small that they would have no essential bearing on those properties, which were important for solving the said technical problem and could be disregarded (see also T.358/04).

In T.643/96 the board held that the concept of bioisosterism did form part of the common general knowledge of those skilled in the art, but that it had to be applied with caution when deciding upon inventive step. In the field of drug design, any structural modification of a pharmacologically active compound was, in the absence of an established correlation between structural features and activity, expected a priori to disturb the pharmacological activity profile of the initial structure. This also held true for an alleged case of bioisosterism, which was one option of a structure-activity relationship, as long as it was not an established case of bioisosterism (see also T.548/91). The board further held that, when deciding upon inventive step in relation to pharmacologically active compounds, what was essential was not whether a particular substructure of a chemical compound was replaced by another known isosteric one, but whether information was available on

the impact of such a replacement on the pharmacological activity profile of the specific (group of) compound(s) concerned (see also T 467/94, T 156/95).

In T 2402/10 the board stated that in the field of drug design, any structural modification of a pharmacologically active compound was, in the absence of an established correlation between structural features and activity, a priori expected to disturb the pharmacological activity profile of the initial structure (see T 643/96, T 548/91).

In T 930/94 the board held that knowledge of the fact that one specific member of a class of chemical compounds did not lead to the effect achieved by several other members of this class, did not, without additional indications, mean that such an effect could be attributed to all the compounds in this group. In such circumstances, the effect in point did not lead to a recognition of the existence of a technical concept that could be generalised (see also T 641/97, T 209/98, T 853/03).

In T 989/93 the board stated that, in the absence of the appropriate common general knowledge, no conclusions were possible on the basis of the known properties of one group of chemical compounds (here: benzene derivatives) regarding the properties of a different group of chemical compounds (here: naphthalene derivatives).

### 9.9.3 Broad claims

In T 694/92 (OJ 1997, 408) the board, with reference to T 939/92 (OJ 1996, 309), explained that Art. 56 EPC 1973 required the claimed invention, i.e. the proposed technical solution for a given technical problem, not to be obvious to a skilled person from the state of the art. If the inventive step of a claimed invention was based on a given technical effect, the latter should, in principle, be achievable over the whole area claimed (see also T 583/93, OJ 1996, 496).

T 939/92 (OJ 1996, 309) contained fundamental rulings on broad claims in the field of chemistry. The board held that in view of the state of the art the technical problem which the patent in suit addressed was the provision of further chemical compounds with herbicidal activity. It was necessary for all the claimed compounds to possess this activity. Moreover, the question as to whether or not such a technical effect was achieved by all the chemical compounds covered by such a claim, might properly arise under Art. 56 EPC 1973, if this technical effect turned out to be the sole reason for the alleged inventiveness of these compounds. The appellants' submission that the test results contained in the description showed that **some** of the claimed compounds were indeed herbicidally active could not be regarded as sufficient evidence to lead to the inference that substantially **all** the claimed compounds possessed this activity. In such a case the burden of proof rested with the appellants. The requirements of Art. 56 EPC 1973 had not therefore been met (T 268/00, T 1188/00, T 320/01, T 1064/01, T 924/02, T 488/16).

Following T 939/92 (OJ 1996, 309), the board stated in T 668/94 that the technical problem could only be taken into account in the assessment of inventive step if it could be accepted as having been successfully solved, i.e. if it were credible that substantially all the claimed compounds possessed the plant growth regulating activity. When only some